WELL WORKOVER PROGRAM

AZADEGAN OIL FIELD

WELL WORKOVER PROGRAM

AZNS-85 (PAD #79), TYPE: IIIF

Drilling Department

JANUARY 2024

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is located in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20^{km} width & 60^{km} length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2^{km} The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643^{m}. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well No. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340^{m} (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 & AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area & south parts of South Azadegan Filed, which was discovered by AZN-1 in crest & AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan & Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus & Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone & partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone & mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province & is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field & in The West of JUFEYR Field. The Average Monthly Temperature is 45$^{°c}$ in summer, & Highest Temperature is 52$^{°c}$. The Lowest Temperature is -4.4$^{°c}$ in winter. Maximum Rainfall is 78^{mm} in February, Winter Season.

This Document covers the work over of well AZNS-85. Main headlines of program are perform 7" tie-back liner lap dry test & complete the well for Fahliyan production. The well was drilled to the final depth of 4090^{m} (Fahliyan Formation) on 2014 by NISOC to produce oil from Fahliyan formation.

1.2 Well Information

Figure 1 illustrates AZNS-85 well location on top of Fahliyan UGC map, nearby wells are AZNS-060 & AZNS-06.

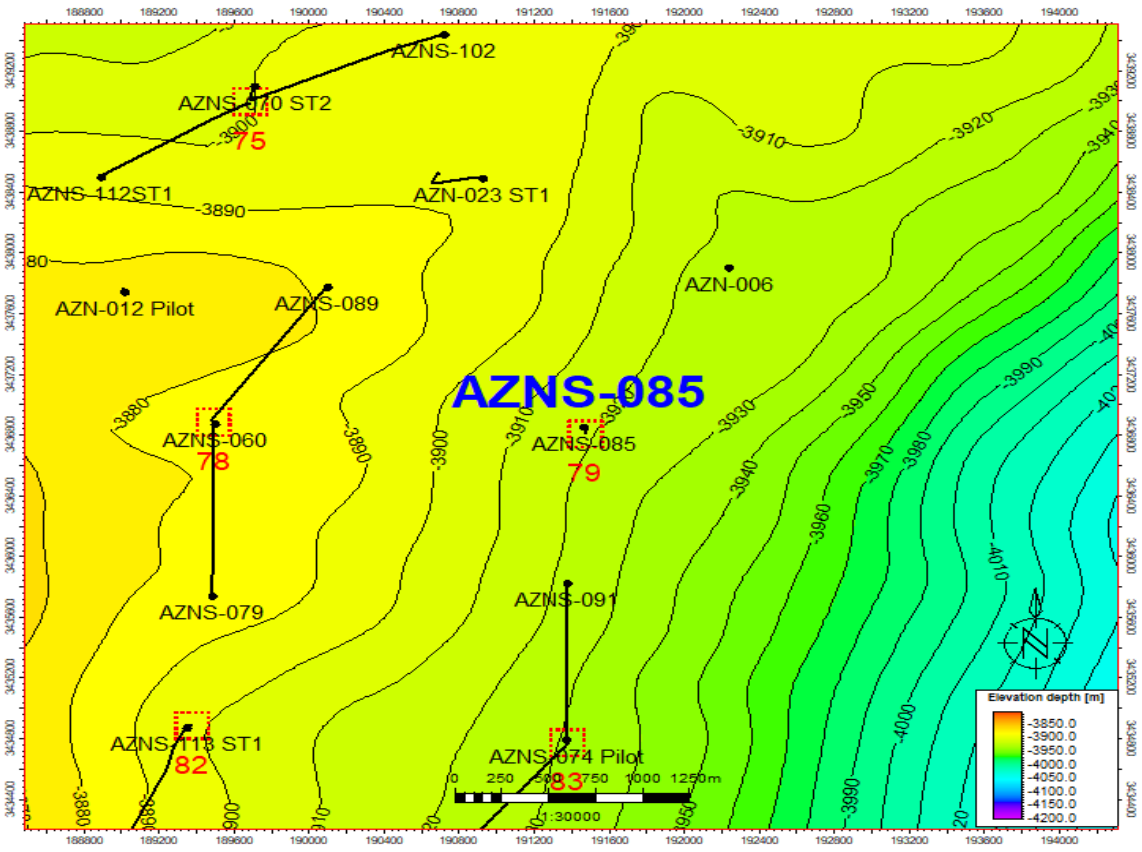


Figure 1: Location of AZNS-85 (Pad 79) on Top of Fahliyan Formation



Figure 2: AZNS Oilfield Location Map

Country / Area Name:	Iran / Khuzestan
Client	Petro Pars Limited
Field Name:	Azadegan
Type of Operation:	Workover
Well Name / Type :	AZNS-85 / Fahliyan well in Block III
Pad Number:	79
Well Type:	Vertical
Wellhead Coordinate:	N: 3,436,812 E: 191,473
TD Coordinate:	N: 3,436,857.89 E: 191,467.47
Well Objective:	Drill CMT plugs, Dry test tied-back liner lap, Rectify the case of any leakage, Perforate 7" liner, Complete the well for Fahliyan production
RTE:	13.45 m
GLE:	3.95 m
Well Duration:	42 (Main Plan) + 5.5 (Contingency Plan)
Total Depth (MD):	4090 m
Producer Formation:	Perforated in 7" liner (Fah2a 4031^{m} – 4045^{m}, Fah3a 4057^{m} – 4061^{m})
Formation TD :	Fahliyan
Reservoir :	Fahliyan

Table 1: AZNS-085 information

1.3 Area Distribution

According to the drilling history, the CONTRACT AREA is distributed into three blocks as per the geological features with regard to DRILLING OPERATION. This distribution may differ according to knowledge accumulated in the future WORK.

Contract area is as per below:

X (m)	Y (m)
185640	3434275
188020	3477361
205224	3476408
202842	3433327

Table 2: Area Distribution

Block I is located on the north part of the CONTRACT AREA in north of well AZN-010. Gachsaran formation is characterized by abnormal high pressure in this block.

Block II is located between well AZN-010 & AZN-011. In this block, Gachsaran formation is characterized by pressure tending to normal.

Block III is located on the south of line well AZN-011. In this block, Gachsaran formation is characterized by normal pressure & all drilled wells in this block suffered from complete loss at Pabdeh formation.

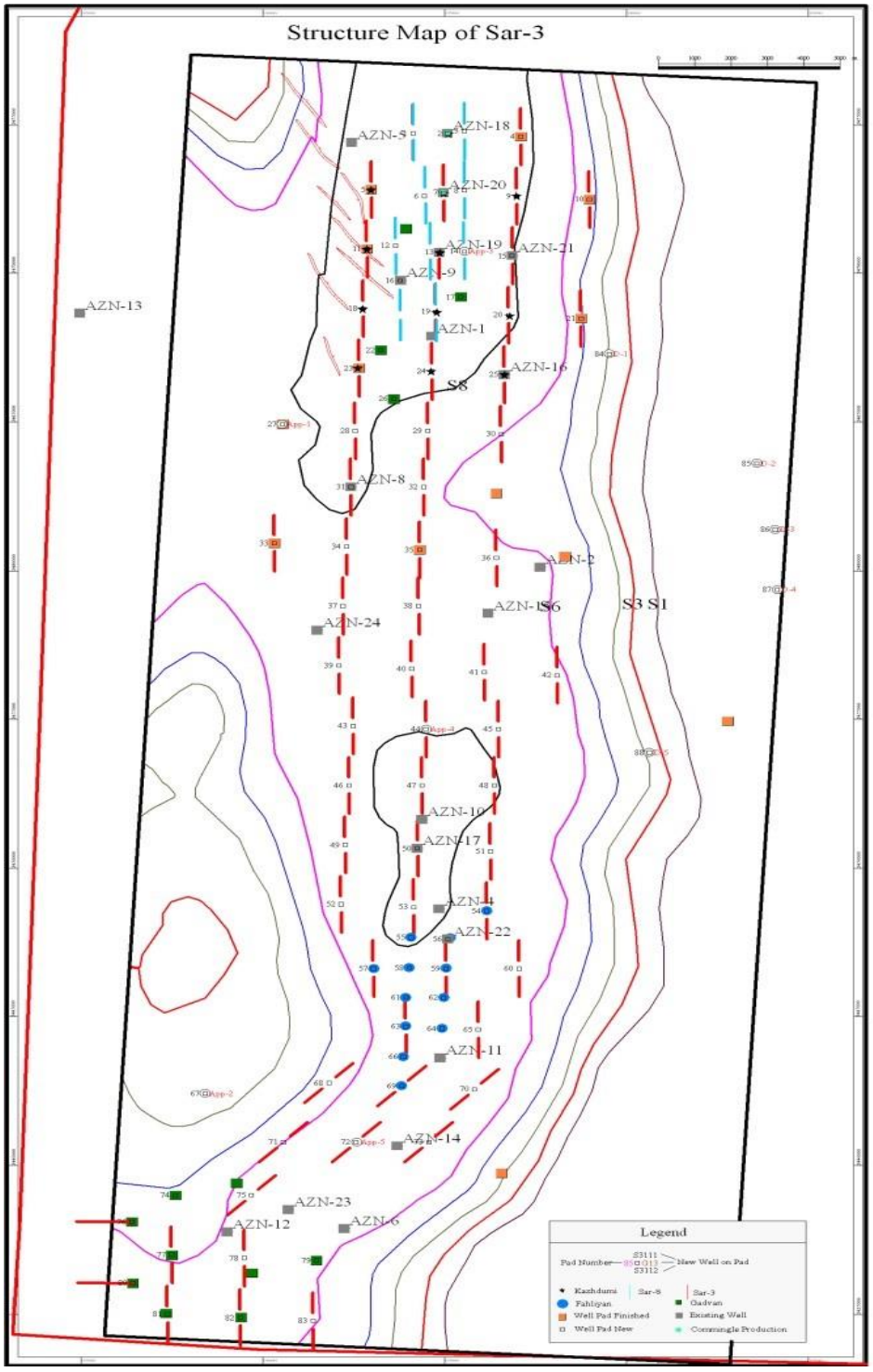


Figure 2: AZNS Area Distribution

1.4 Well Pad Specification

Well Pad Type	Well Pad Dimension (m)	Mud Pit Dimension (m)	Cellar Dimension (m)	Conductor Size
Single Well	130 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D	30"
Double Well	145 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D	30"
Triple Well	160 x 100	25 x 20 x 2	7.5L x 3.80W x 3.40D + 4.0L x 3.8W x 3.4D + 4.0L x 3.8W x 3.4D	30'

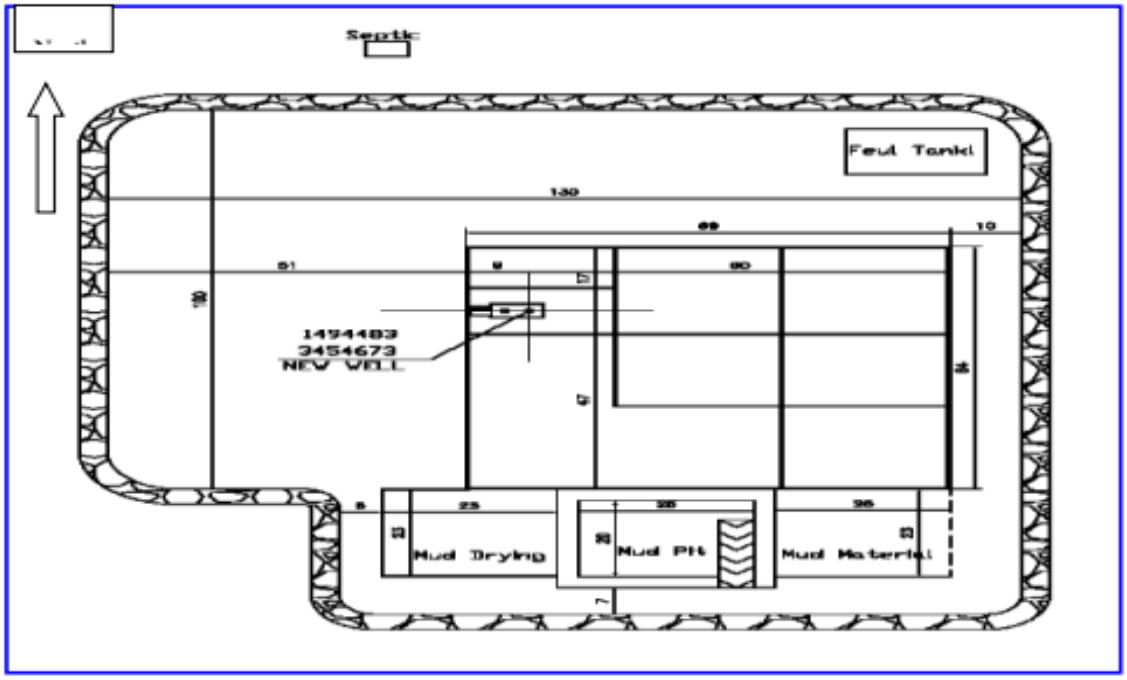
Table 3: Well Pad Specification

Compressive Strength for reinforced concrete after 28 days: 280 $^{kg/cm2}$

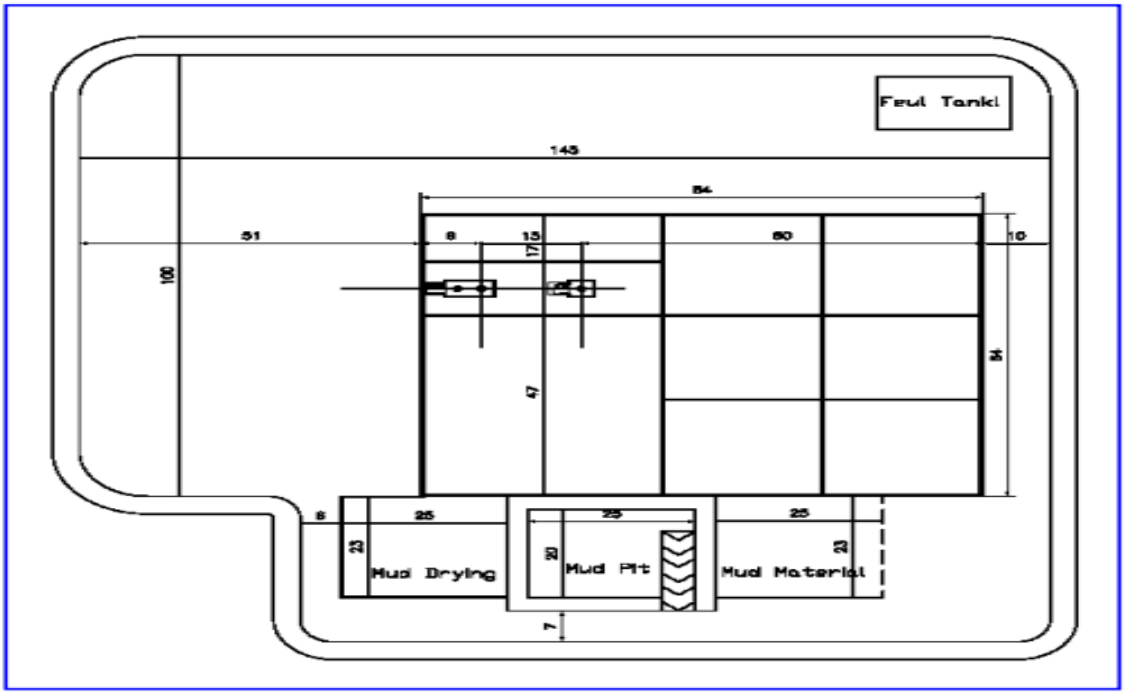
- Fuel tank concrete pond dimension: Length 20^{m}×Width 12^{m}×Height 0.5^{m}
- Drainage: Width 30^{cm} × Depth 30^{cm}
- Burning pit: 15^{m}×15^{m} square, located on South-East or South-West with at least 350^{m} distance from well head
- Cellar to cellar distance in multi-well well pad: 15^{m}
- Conductor pipe depth under cellar bottom is 5.5^{m}. The conductor pipe will be driven while constructing the well pad.
- Mud material dimension is 25^{m}*35^{m} +granular surface
- Mud drying dimension is 25^{m}*35^{m} + concrete surface
- Well pad surface is gravel. Access road granular surface. The level of well pads is 3.95 MSL

1. Types of well pads:

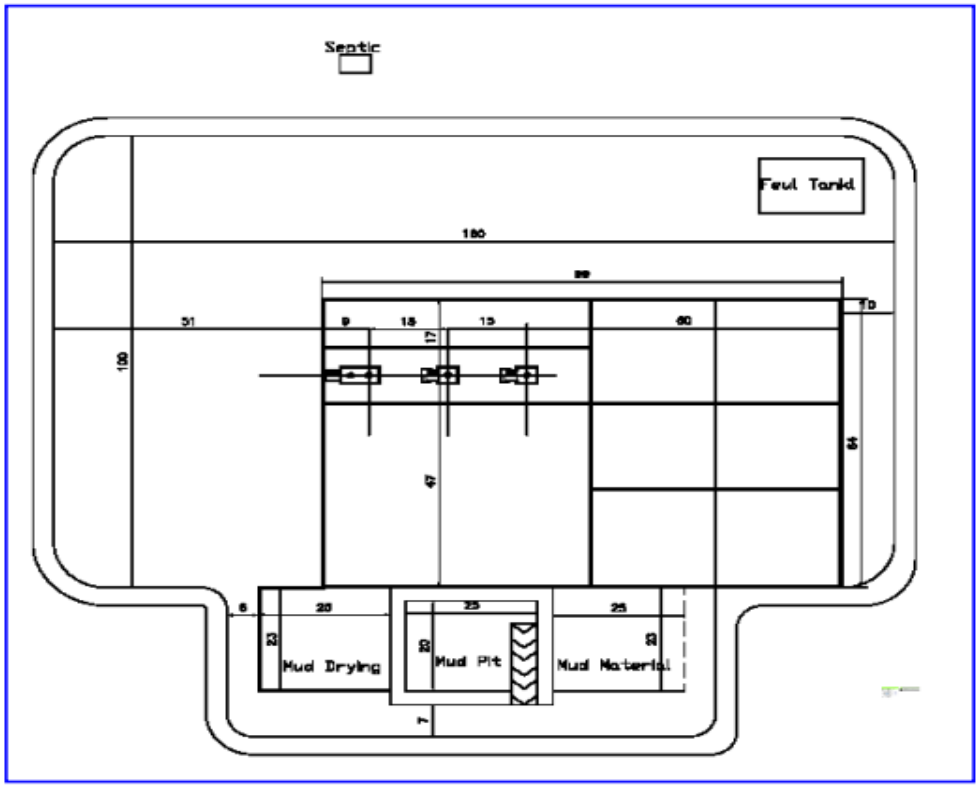
- One well in one pad.
- Two wells in one pad. 2 Sarvak horizontal wells or 1 Sarvak horizontal well plus 1 vertical/directional well).
- Three wells in one pad: 2 Sarvak horizontal wells & 1 vertical/directional wells.



Well Pad for one well



Double-well Well Pad



Triple-well Well Pad

Figure 3: Well Pads Sketch

1.5 Reservoir Data

Reservoir Name:	Fahliyan	Unit
Fluids:	Oil	….
Datum Depth:	-4000	M,TVDss
Datum Pressure:	Fah-2a= max 8400 psi Fah-3a= max 8600 psi	psi
Pressure Gradient:	0.28	Psi/ft.
Temperature:	132	C°
Temperature Gradient:	1.32 F/100ft 0.73 C/100ft 2.4 C/100m	F/100ft C/100ft C/100m
H_2S:	H2S in Gas at Atm condition 3000 PPM	Ppm
CO_2:	Co2 in Gas at Atm Condition 50000 PPM	Ppm
Salt Salinity :	143,000	ppm
Calcium Content:	8,750	(mg/lit)

Table 4: Reservoir Data

1.6 Predicted Pressure, Temperature and OWC

Predicted Pressure and Temperature and TTZ Contact								
Reservoir	**Sub-layer**	**Mid of Sub-Layer Depth in AZNS-085 (mss)**	**Temp (°F)**	**Temperature Gradient (°F/100ft)**	**Current Reservoir Pressure (psia)**	**Basis for Reservoir Pressure**	**Top of Transition Zone (mTVDSS)**	**Basis for TTZ and FWL**
Fahliyan	Fah-1	-3988	257	1.32	9252	Pressure Gradient	-4149	Dynamic Model
	Fah-2a	-4026	259		8030		-4045.6	
	Fah-2b	-4039	259		8045		-	
	Fah-3a	-4047	260		8350		-4151	
	Fah-3b	-4056	260		8362		-	
	Fah-4a	-4066	261		7752		-4057.8	
	Fah-4b	-4073	261		7763		-4058.4	

1.7 Formation Pressure Prediction in Block III

Predicted Formation Pressure in Block III				
Formation	**Interval**	**Pore Pressure**	**Collapse Pressure (Break-Out Pressure)**	**Fracture Pressure (Break-Down Pressure)**
	m	*SG*	*SG*	*SG*
Aghajari	0-973	0.98-1.18	0.06-1.39	1.01-1.89
Gachsaran	973-1279	1.13-1.31	0.78-1.36	1.89-2.01
Asmari	1279-1473	1.14-1.26	0.6-1.28	1.43-1.79
Pabdeh	1473-2123	1.20-1.26	0.82-1.36	1.40-1.78
Gurpi	2123-2538	1.21-1.26	0.66-1.02	1.40-1.76
Ilam	2538-2612	1.18-1.24	0.20-0.9	1.41-1.72
Laffan	2612-2619	1.13-1.18	0.24-1.11	1.41-1.72
Sarvak	2619-3260	1.14-1.26	0.24-1.23	1.44-1.81
Kazhdumi	3260-3463	1.1-1.14	0.46-0.96	1.93-1.94
Dariyan	3463-3626	1.18-1.25	0.55-1.1	1.96-2.06
Gadvan	3626-3932	1.1-1.25	0.55-1.44	1.97-2.11
Fahliyan	3932-T.D.	1.60-1.3	1.20-1.75	2.08-2.12

Table 5: Formation Pressure Prediction

1.8 Drilling History

The AZNS-85 (Pad 79) was drilled by rig 86 Fath, which rig moving was spudded on 15-07-1394 @ 00:00.

- 26" hole section drilling to 152^{m}, run 20" casing to 150^{m} (J-55, 133#, BTC) & cement job as program with 60^{bbl} cement return, Mw 68^{pcf}.
- 17-1/2" hole was drilled to 1279^{m}, while increase mud weight f/67 to 85^{pcf}, run 13-3/8" casing (N-80, BTS, 72#, f/0 to 1099.15^{m} & NKT, BTS, 72#, f/1099.5^{m} to 1279^{m}) & cement job as program with 180^{bbl} cement return.
- 12-1/4" hole section was drilled to 3237^{m} with 6-150^{BPH} loss while pumping LCM Pill & bypass system, take full set log, continue drilling to 3367^{m} with 20-150^{BPH} loss while pumping LCM Pill & bypass system, complete loss pump LCM Pill several time, decrease Mw f/80 to 77^{pcf}, hole condition & While condition stuck string & complete loss, work on stuck & free pipes, pump LCM Pill & increase Mw to 70^{pcf}, drilling to 3361^{m}, flow the well, twist-off string, fish left hole f/3029.37^{m} to 3496.85^{m} (12-1/4" PDC bit, bit sub, 9 Jts D.C, 8" Jar, 3 Jts 8-1/2" D.C, XOS, 36 Jts 5" DPs), fish recovered, increase Mw to 78^{pcf}, drilling to 3950^{m}, take log as program, run 9-5/8" casing (L-80, 53.5#, VAM f/0 to 1844^{m} & 110SS, 47#, TPCQ f/1844^{m} to 3950^{m}) & cement job as program, no cement return.
- The hole section 8-1/2" was drilled to 3957^{m}, take CBL/VDL/CCL/GR log from 9-5/8" casing, drilling to 4034^{m}, increase Mw to 101^{pcf} due to flow well, drilling to 4090^{m} (TD), suddenly well flow (20^{BPH} total gain 30^{BPH} oil & gas in mud; 92^{pcf}) increasing Mw to 103^{pcf}, run 7" liner (VM110SS, 35#, N.VAM) to 4090^{m}, pressure test & inject 150^{bbl} cement in 9-5/8" casing @ 3103^{m}, dry test 7" liner lap was OK, take CBL/VDL/CCL/GR log from 7" liner, set cement plug f/3882^{m} to 3650^{m} & test 9-5/8" casing was not OK, take CAST log f/3648^{m} to 1200^{m} in 9-5/8" casing, observe tear 9-5/8" casing f/3157^{m} to 3140^{m}, first tie back 7" liner to 3012^{m} (L-80, 29#, HSM1), dry test 7" liner lap was not OK, second tie back 7" liner lap to 2903^{m} (L-80, 29#, VAM), set cement plugs f/3879^{m} to 3663^{m} & f/2903^{m} to 2763^{m}, suspension well without completion string & release rig @ 1395-02-27.

1.9 Casing & Liner Depth

Casing Spec.	20"	J-55, 133#, BTS	0 - 150^{m}
	13-3/8"	N-80, 72#, BTS NKT-95, 72#, BTS	0 - 1098^{m} 1098^{m} – 1277^{m}
	9-5/8"	TN110SS, 47#, TPCQ L-80, 53.5#, VAM	0 – 1844^{m} 1844^{m} – 3950^{m}
	7" Tie back #1 Tie back #2	VM110HC-SS, 35#, N.VAM L-80, 29#, HSM1 L-80, 29#, VAM	3829^{m} – 4090^{m} 3012^{m} – 3829^{m} 2903^{m} – 3012^{m}

Table 6: Casing & Liner Data

1.10 Well Schematic before Workover

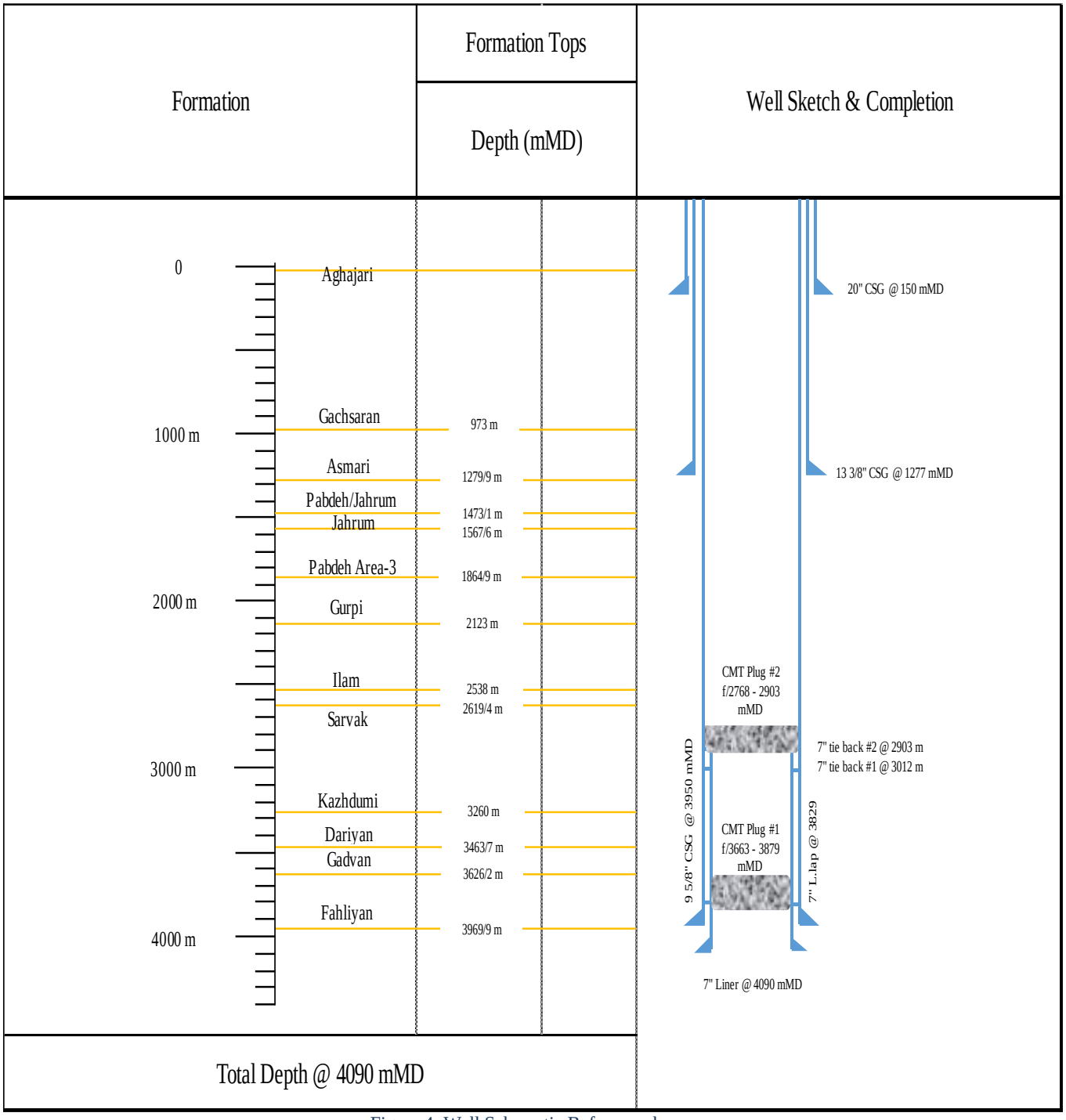


Figure 4: Well Schematic Before workover

1.11 Well Schematic after Workover

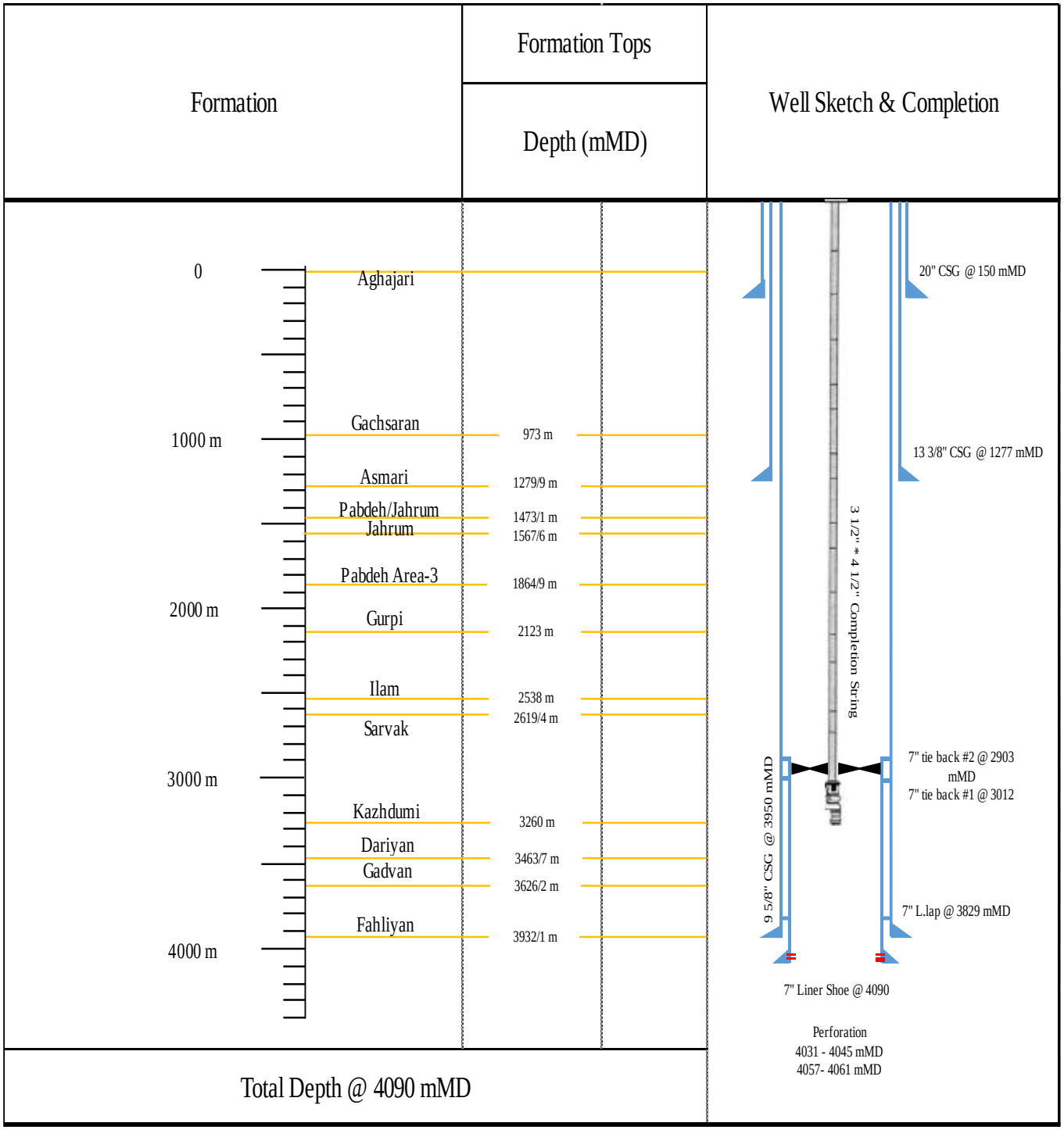


Figure 5: Well Schematic After workover

2. Geological Information

2.1 Formation Stratigraphy & Lithology

Stage	Group	Formation	Lithology	Lithology Description	Block I Top (mTVD)	Block II Top (mTVD)	Block III Top (mTVD)
Pliocene Miocene	Fars	Agha Jari		Marl, Sandstone, Anhydrite	Surface	Surface	Surface
Miocene	Fars	Gachsaran		Marl, Anhydrite, Claystone	1380	1150	961
E. Miocene Oligocene		Asmari		Limestone, Sandstone, Claystone	1776	1503	1318
E. Oligocene L. Paleocene		Pabdeh		Limestone, Argillaceous Limestone, Claystone, Dolomite	2079	1786	1525
Maastrichtian Campanian		U. Gurpi / Tarbur / L. Gurpi		Limestone, Argillaceous Limestone	2352	2303	2153
Santonian Coniacian	Bangestan	Ilam		Limestone, Claystone	2606	2534	2549
	Bangestan	Laffan		Claystone	2691	2642	2625
Turonian Cenomanian	Bangestan	Sarvak		Limestone, Claystone	2697	2648	2632
Cenomanian Albian	Bangestan	Kazhdumi		Sandstone, Argillaceous Limestone, Claystone, Shale	3335	3300	3269
Aptian	Khami	Dariyan		Limestone	3538	3499	3471
Aptian Barremian	Khami	Gadvan		Argillaceous Limestone, Claystone, Sandstone	3718	3696	3635
Neocomian	Khami	Fahliyan		Limestone, Marl	4092	3989	3974

Figure 6: Azadegan Stratigraphic Column

3. Logging Program

Hole Size	Casing Size	Logs
		Cased Hole
12-1/4"	9-5/8"	SBT/VDL/CCL/GR Corrosion Log
8-1/2"	7"	SBT/VDL/CCL/GR Corrosion Log CCL after perforation

Table 7: Logging Program

4. Program Outline

4.1 Contractors Involved

Table 8: Contractor Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization & Demobilization the Rig, Rig Move Between Well Pads	OEOC 201
2	Drilling Fluids & Completion Fluids	VIRA
3	Waste Management	Pars Hafaran Arvand
4	Electrical Logging & Perforation	SEALAND
5	Mud Logging	MAPSA KISH
6	H2S Monitoring & Control	MAPSA KISH BONYAN IMEN
7	Casing & Tubing Running	OEOC
8	Cementing	SPAD ENERGY
9	Dry Test, DST & RTTS	OEOC
10	Slick Line	OEOC
11	Stimulation (Coiled Tubing & Acidizing & Nitrogen Lifting)	OEOC
12	Surface Well Testing	OEOC
13	Liner Hanger	PARS POOLAD
14	Well Head	OEOC

4.2 Time Breakdown

De-Complete, Work-over and Completion Operation TBD of Well: AZNS-85 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Work-over									
Held PJSM. Spud the Rig. Stand Back required 5" DPs, 5" HWDPs and 6-1/2" DC.	SPUD	2763	30.0	1.25	1.3	1.4	33.0	1.38	1.38
Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status. Note: If wellhead pressure is not ZERO after bleed off, contractor shall inform client. Remedial plan to be prepared in advance by operators & will be followed after client's confirmation.	Wellhead	2763	6.0	0.25	1.5	0.3	6.6	0.28	1.7
N/D X-Mass Tree	BOP	2763	4	0.17	1.7	0.2	4.4	0.18	1.8
N/U BOP and pressure test.	BOP	2763	16	0.67	2.3	0.7	17.6	0.73	2.6
RIH with tubing hanger retrieving tool & engage with TBH. Pull out with TBH to surface.	Trip	2763	4	0.17	2.5	0.2	4.4	0.18	2.8
Perform wellhead pressure integrity test with cup tester.	Trip	2763	4	0.17	2.7	0.2	4.4	0.18	2.9
RIH with 8-1/2" D/O BHA to top of cement plug.	Trip	2763	10	0.42	3.1	0.5	11.0	0.46	3.4
Drill out cement plug to 7" liner lap f/2768m to 2903m (take care of lap@ 2903m), POOH.	D/O	2903	48	2.00	5.1	2.2	52.8	2.20	5.6
RIH with 5-7/8" clean-out BHA including 9-5/8" scraper , scrape dry test packer setting depths, continue RIH to top of cement @ 3663m & tag, CBU till shakers clean, POOH.	D/O	3663	36	1.50	6.6	1.7	39.6	1.65	7.2
RIH with 9-5/8" RTTS packer & perform 7" liner lap dry test, POOH. Note: If dry test is not OK, follow contingency plan #1.	Trip	3663	60	2.50	9.1	2.8	66.0	2.75	10.0
Perform BOP test.	BOP Test	3663	6	0.25	9.3	0.3	6.6	0.28	10.3
RIH with 5-7/8" D/O BHA (3-1/2" DPs JT by JT), drill-out cement plug inside 7" liner f/3663m to 3879m, continue RIH to 4070m LDC, CBU till shakers clean, POOH to surface.	D/O	4070	60	2.50	11.8	2.8	66.0	2.75	13.0
RIH with tandem scraper 9-5/8" * 7", scrape inside 9-5/8" casing & 7" liner (dry test packer setting depths), CBU & POOH.	Trip	3663	40	1.67	13.5	1.8	44.0	1.83	14.9
RIH with wireline SBT/VDL/CCL/GR Logging tools to bottom, log up to surface (in two separate runs for 9-5/8" CSG & 7" Liner), POOH.	Log	3663	48	2.00	15.5	2.2	52.8	2.20	17.1
RIH with wireline corrosion logging tools to bottom, log up to surface (in two separate runs for 9-5/8" CSG & 7" Liner), POOH.	Log	3663	48	2.00	17.5	2.2	52.8	2.20	19.3
Perform BOP test.	BOP Test	3663	6	0.25	17.8	0.3	6.6	0.28	19.5
RIH with 7" scraper, scrape inside 7" liner, CBU till shakers clean, displace inside 7" liner with non-solid fliud, POOH to surface.	Completion	4070	40	1.67	19.4	1.8	44.0	1.83	21.4
Run perforation gun assembly & perforate 7" liner in Fah2a (4031m to 4045m) & Fah3a (4057m to 4061m) tentatively, POOH to surface.	Completion	4070	60	2.50	21.9	2.8	66.0	2.75	24.1
RIH with CCL log to confirm perforation intervals.	Log	4070	12	0.50	22.4	0.6	13.2	0.55	24.7
Perform BOP test.	BOP Test	4070	6	0.25	22.7	0.3	6.6	0.28	24.9
RIH with 7" scraper to scrape completion packer area & displace the well with non-solid as completion fluid, POOH to surface & L/D DP's & DC's.	Completion	4070	60	2.50	25.2	2.8	66.0	2.75	27.7
Run 10K completion string as per program.	Completion	4070	96	4.00	29.2	4.4	105.6	4.40	32.1

Set TWCV inside TBH, N/D BOP & install 10K XMT, pressure test same.	Completion	4070	36	1.50	30.7	1.7	39.6	1.65	33.7
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion	4070	5	0.21	30.9	0.2	5.5	0.23	34.0
Retrieve BPV, RIH with SL, set protection sleeve, retrieve Plug & Prong from NO-GO landing nipple, POOH.	Completion	4070	60	2.50	33.4	2.8	66.0	2.75	36.7
RIH with CT to bottom, displace the well with diesel, POOH.	Completion	4070	12	0.50	33.9	0.6	13.2	0.55	37.3
R/U & RIH with SL, retrieve protection sleeve, POOH & R/D SL.	Completion	4070	6	0.25	34.1	0.3	6.6	0.28	37.5
Flow the well as pre-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.	Completion	4070	72	3.00	37.1	3.3	79.2	3.30	40.8
R/U & RIH with SL, set protection sleeve, POOH & R/D SL.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
RIH with CT & perform acid wash as per program, POOH to surface.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
R/U & RIH with SL, retrieve protection sleeve, POOH & R/D SL.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for minimum 96 hrs. Note: The time is only for flowing duration (R/U, R/D time & preparation is excluded. Note: Acidizing & post-acid clean-up maybe perform as rigless job upon contractor & client approval but in case of any failure in well integrity, contractor should provide drilling rig & required services in own responsibility & fix the problem.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
RIH with SL, set protection sleeve.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
RIH with CT to bottom & displace the well with diesel, POOH.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
RIH with SL, install Plug & Prong into NO-Go landing nipple, POOH.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
RIH with SL, retrieve protection sleeve.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.	Completion	4070	Rig Less	0.00	37.1	0.0	0.0	0.00	40.8
Total Duration of Work-over:				***37.1***	***37.1***			***40.8***	***40.8***
NOTE:									
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip	4070	4	0.17	37.3	0.2	4.4	0.18	41.0
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service	4070	21	0.88	38.2	1.0	23.1	0.96	42.0
Total Duration of AZNS-85 Work-over Well:		**4392**			**38.2**				**42.0**
Contingency Plans TBD of Well: AZNS-85 Work-over	IADC Code	Depth (meters)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
Based on operation situation in each step, the following contingency plans might be followed									
contingency plan #1: Install liner SN-packer on top of 7" liner lap and set completion packer above 7" lower Liner lap@ 3012m.	Trip	3663	120	5.00	5.0	5.50	132	5.50	5.5
Total Duration of AZNS-85 Work-over Well:		**4070**			**5.0**				**5.5**

5. Sequence of Operation

5.1 Main Objective

Base on well condition, the decision has been made to perform 7" tied-back liner lap dry test &

Main plan

1. Held PJSM. Spud the Rig. Stand back required 5" DPs, 5" HWDPs and 6-1/2" DC.

2. Check the wellhead for any damage or leakage. Record well head pressures; check the all annulus pressure. Check the return fluid & Bleed off any annulus pressure to Zero upon client's approval and recheck the well status.

Note: If wellhead pressure is not ZERO after bleed off, contractor shall inform client. Remedial plan to be prepared in advance by operators & will be followed after client's confirmation.

3. N/D X-Mass Tree.
4. N/U BOP and Pressure Test.
5. RIH with tubing hanger retrieving tool & engage with TBH. Pull out with TBH to surface.
6. Perform wellhead pressure integrity test with cup tester.
7. RIH with 8-1/2" D/O BHA to top of cement plug.
8. Drill out cement plug to 7" liner lap f/2768m to 2903m (take care of lap@ 2903m), POOH.
9. RIH with 5-7/8" clean-out BHA including 9-5/8" scraper , scrape dry test packer setting depths, continue RIH to top of cement @ 3663m & tag, CBU till shakers clean, POOH.
10. RIH with 9-5/8" RTTS packer & perform 7" liner lap dry test, POOH.

Note: If dry test is not OK, follow contingency plan #1.

11. RIH with 5-7/8" D/O BHA (3-1/2" DPs JT by JT), drill-out cement plug inside 7" liner f/3663m to 3879m, continue RIH to 4070m LDC, CBU till shakers clean, POOH to surface.
12. RIH with tandem scraper 9-5/8" * 7", scrape inside 9-5/8" casing & 7" liner (dry test packer setting depths), CBU & POOH.
13. RIH with wireline SBT/VDL/CCL/GR Logging tools to bottom, log up to surface (in two separate runs for 9-5/8" CSG & 7" Liner), POOH.
14. RIH with wireline corrosion logging tools to bottom, log up to surface (in two separate runs for 9-5/8" CSG & 7" Liner), POOH.
15. RIH with 7" scraper, scrape inside 7" liner, CBU till shakers clean, displace inside 7" liner with non-solid fluid, POOH to surface.
16. Run perforation gun assembly & perforate 7" liner in Fah-2a (4031m to 4045m) & Fah-3a (4057m to 4061m) tentatively, POOH to surface.
17. RIH with CCL log to confirm perforation intervals.

18. RIH with 7" scraper to scrape completion packer area & displace the well with non-solid as completion fluid, POOH to surface & L/D DP's & DC's.
19. Run 10K completion string as per program.
20. Set TWCV inside TBH, N/D BOP & install 10K XMT, pressure test same.
21. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, and Release the Rig.
22. Retrieve BPV, RIH with SL, set protection sleeve, retrieve Plug & Prong from NO-GO landing nipple, POOH.
23. RIH with CT to bottom, displace the well with diesel, POOH.
24. R/U & RIH with SL, retrieve protection sleeve, POOH & R/D SL.
25. Flow the well as pre-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for 72 hrs.
26. R/U & RIH with SL, set protection sleeve, POOH & R/D SL.
27. RIH with CT & perform acid wash as per program, POOH to surface.
28. R/U & RIH with SL, retrieve protection sleeve, POOH & R/D SL.
29. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for minimum 96 hrs.

Note: The time is only for flowing duration (R/U, R/D time & preparation is excluded.
Note: Acidizing & post-acid clean-up maybe perform as rigless job upon contractor & client approval but in case of any failure in well integrity, contractor should provide drilling rig & required services in own responsibility & fix the problem.

30. Flow the well as post-acid clean-up till acceptable BSW & reading stable flow rate on 2 different choke size for minimum 96 hrs.
31. RIH with SL, set protection sleeve.
32. RIH with CT to bottom & displace the well with diesel, POOH.
33. RIH with SL, install Plug & Prong into NO-Go landing nipple, POOH.
34. RIH with SL, retrieve protection sleeve.
35. Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV.

5.2 Contingency Plan

Based on operation situation in each step, the following contingency plans might be followed.

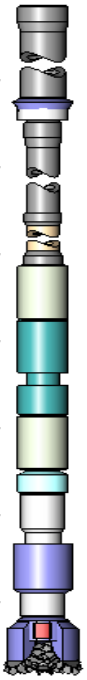
5.2.1 Contingency Plan #1

Install liner SN-packer on top of 7" liner lap and set completion packer above 7" lower Liner lap@ 3012m.

6. BHA Program

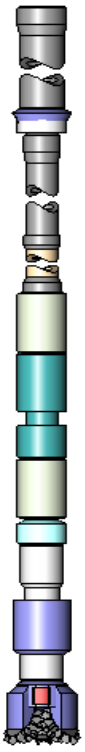
6.1 8-1/2" Drill-Out BHA

BIT SIZE (inch)	8-1/2"	BHA Configuration to D/Out Inside 9-5/8" Casing	Collar / OD
Depth In/Out (m)	2763 - 2903		DC-1 / 6-1/2"
HOLE INCLINATION	0		
MUD WEIGHT (pcf)	106		
B.FACTOR	0.784		
Weight in Air (lb)	24,624		
Bouyed Weight (lb)	19,297		
Bouyed Weight Below Jar (lb)	14,267		

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	3	5	3	4-1/2" IF Pin x Box	27.42	182.71
	XOS	1	6-1/2	3	4-1/2" Reg. P x 4-1/2" IF B	1	1
	6-1/2" Drill Collar	3	6-1/2	2-13/16	4-1/2" Reg. P x B	27.42	155.29
	6-1/2" Jar	1	6-1/2	2.75	4-1/2" Reg. P x B	8	127.87
	6-1/2" Drill Collar	13	6-1/2	2-13/16	4-1/2" Reg. P x B	118.82	119.87
	6-1/2" Bit Sub W/ Float Valve	1	6-3/4"	3	4-1/2" Reg. B x B	0.75	1.05
	8-1/2" MT Bit	1	8-1/2"	-	3-1/2" Reg. P x B	0.3	0.3

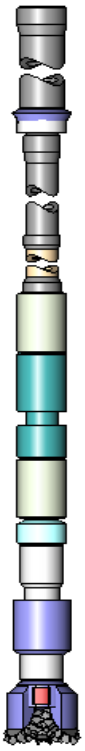
6.2 5-7/8" Drill-Out BHA

BIT SIZE (inch)	5-7/8"	BHA Configuration to D/Out Inside 7" Liner	Collar	OD
Depth In/Out (m)	3663 - 3879		DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,229			
Bouyed Weight Below Jar (lb)	16,462			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	5-7/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

6.3 5-7/8" Clean-Out BHA

BIT SIZE (inch)	5-7/8"	BHA Configuration to C/Out Inside 7" Liner	Collar	OD
Depth In/Out (m)	2903		DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,229			
Bouyed Weight Below Jar (lb)	16,462			

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	344.61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115.2	343.61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54.84	228.41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27.42	173.57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	146.15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137.1	138.15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0.75	1.05
	5-7/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0.3	0.3

7. Mud & Cement Program

Note: Detailed mud & cement program, will be issued in due time as per client approved template.

7.1 Workover Hole Section

Workover section	7" , 4 ½" Liner
From: 1699m	TO: 4090m
Kill the well, Drill out CMT plugs, Complete the hole with Completion 10K & Completion Fluid	

7.1.1 Overview

Drilling Fluid System	Kill Mud & Completion Fluid
Key products	Workover Fluid , Completion Fluid, Kill Mud
Water type	Drill Water
Solid Control	Shale Shaker
Potential problems	Down Hole losses , gas/oil influx, H_2S,

Interval Mud Properties						
Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
0-4090	106	30-55	8-22	NC	9-10.5	45-55

Interval Completion Fluid Properties						
Depth interval (m)	MW (pcf)	Plastic viscosity (cp)	Yield point (lb/100ft²)	API Fluid Loss (cc/30 Min)	pH	Funnel Viscosity (sec/qt)
Surface~4090	106	ALAP	N/A	NC	6-7.5	27-30

7.1.2 Mud Discussion & Mixing Procedure

7.1.2.1 Mud type

Kill mud 106^{pcf} will be used to kill the well and well will be displaced by Completion fluid of $CaBr_2$ 106^{pcf}. Attention must be paid that there is Calcium Bromide Brine inside the well and all of them must be retrieved in separated mud tank by the time of displacing the well by kill mud. (As what the hole dictates, mud properties may be varied by the time).

7.1.2.2 Drilling Fluid Formulation

Kill Mud/Work over Mud formulation		
Component	**Conc. (ppb)**	**Component Description**
Drill water	0.5766 bbl/bbl	Base Fluid
Caustic Soda	0.3	PH Modifier (lime can be used instead)
Soda Ash	0.3	Hardness Remover and
Sodium Chloride	63	Non-Solid weighting Agent
Xanthan-Gum	0.2	Viscosifier (In case of need and based on situation can be used)
Calcium Carbonate	325	Weight Agent

*Above calculation is based on mud weight 106^{pcf}

*Lime can be substituted for caustic soda

Completion Fluid Formulation		
Component	**Conc. (ppb)**	**Component Description**
Calcium Bromide (14.2^{ppg})	0.984 bbl/bbl	Base Fluid

Completion Fluid Formulation		
Component	**Conc. (ppb)**	**Component Description**
Corrosion Inhibitor	1% Vol	Anti-Corrosion
H_2S Scavenger Liquid	0.5% Vol	H2S Removal/ Anti Corrosion
Oxygen Scavenger CA	0.5 ppb	Oxygen Removal/Anto Corrosion

7.1.2.3 Drilling Fluid Formulation

Kill Mud Mixing Procedure:

1. Fill tanks with calculated amount of Drill Water $0.576^{bbl./bbl.}$
2. Treat Drill Water with Caustic soda and Soda Ash by 0.3^{pbb}
3. Mix NaCl at 66^{ppb}.
4. Add Xanthan-Gum by 0.2^{pbb}.
5. Add $CaCO_3$-F (325^{ppb}) to reach MW of $\pm 106^{pcf}$

   ↳ In case of foaming use Alcohol-Based Defoamer.

Note: Use Hi Vis pill to clean hole if necessary.
Calculation is based on per bbl. of mud.

Completion Fluid Mixing Procedure:

1. Make sure that all pits and lines involved with completion fluid are thoroughly cleaned, flushed and free of any fluid and solid.

   ↳ Make sure that at least one large pit is kept empty for transferring ready brine into after given time for settling. After transferring one pit to empty pit, the first one is to be flushed and cleaned for accommodating the brine of next ready pit, and so on to the last pit.

   ↳ Depends on the rig facilities and structure, mud engineer on board may find it better to prepare the fluids pit-by-pit.

2. Transfer the content of $CaBr_2$ drums into designated pits.

   ↳ Record the number of drums poured in each tank for accurate volume in it as the base of calculation for adding other additives into; do not rely on pit gauge, sounding or mud logging sensor, but have a look on them for comparison and/or correlation if found useful.

3. Mix Corrosion Inhibitor into pits at concentration of 3.5^{pbb} (1% by volume).
4. Mix H_2S Scavenger Liquid into pits at concentration of 0.5% by Volume.
5. Do not add Oxygen Scavenger CA at this stage as it should be added directly from the top of the pit, just prior to pump the fluid into the well.
6. Shut down all agitators and mixing pumps, and leave the mixed fluid in the pits for at least 12 hours to let any solid particle inside (mostly insoluble Lime) settles down.
7. Start transferring the clear fluid of the first pit to the empty pit by means of an air-powered Wilden pump.

   ↳ Make sure that the suction hose of the Wilden pump is not put deep close to the bottom of the pit, otherwise, settled solids are sucked in.

8. After finishing the first pit, stop Wilden pump, wash, flush and clean the pit, use it as the empty pit for accommodating the fluid of second pit using the and same above method.

Note: all $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.
Based on well readings and the operation, mud weight may be varied.

7.1.2.4 Downhole Losses

As perforation had been done in well AZN#85 & there is exposure to reservoir, risk of any probable formation losses must be taken into consideration.
Losses will be minimized/avoided by the planned use of properly sized bridging/weight material (and Polymer) in conjunction with appropriate drilling practices. If seepage losses are noted on the well no specific action will be taken but close monitoring of the loss rate is to be made. For moderate loss rates addition of $CaCO_3$ 50 micron and 150 micron is recommended to ensure that there is adequate correctly sized material to act as bridging agents.

Seepage losses Less than 10 $^{bbl./hr}$.

No specific action will be taken, but close monitoring of the probable loss rate is made.

Losses from 10-30 $^{bbl./hr}$.

- Load $CaCO_3$-M (150-300 micron) in system up to 5^{ppb}

***If loss not cured and even tends to increase, then:**

- Pump 40-60^{bbl} of pill made by 7-10^{ppb} $CaCO_3$ 150-300µ & 7-10^{ppb} $CaCO_3$ 600µ

Losses from 30-50 $^{bbl./hr}$.

- Load $CaCO_3$-M (150-300 micron) in system up to 5^{ppb}
- Since some of the $CaCO_3$-M will be discarded over the shakers, $CaCO_3$-M to compensate the

separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60^{bbl} of pill made by
- 10-20^{ppb} $CaCO_3$ 200-300µ
- 10-20^{ppb} $CaCO_3$ 600 µ
- 10^{pbb} Oyster Shell-F
- 10^{pbb} Oyster Shell-M.

Losses from 50-100 $^{bbl/hr}$.

- Load $CaCO_3$-M (150-300 micron) in system up to 5^{ppb}
- Load $CaCO_3$-C (600-100 micron) in system up to 5^{ppb}
- Since some of the $CaCO_3$-M&C will be discarded over the shakers, add $CaCO_3$-M&C to compensate the separated amount on the shakers.

***If loss not cured and even tends to increase, then:**

- Pump 40-60^{bbl} of the following pill:

- 10-20^{ppb} $CaCO_3$ 200-300 micron
- 10-20^{ppb} $CaCO_3$ 600 micron
- 7-10^{ppb} of each $CaCO_3$ 1000 micron & $CaCO_3$ 2000 micron
- 5-10^{pbb} Oyster Shell -F
- 5-10^{pbb} Oyster Shell -M

Losses more than 100 $^{bbl/hr}$.

High concentrated LCM pill of various sizes of Calcium Carbonate and Oyster Shell-F&M&C should be pumped and spotted and gently squeezed over the loss zone.

Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30 minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close BOP and squeeze 10^{bbl} LCM pill into formation without exceeding 200^{psi}.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50^{bbl} of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20^{bbl} of LCM pill into the formation without exceeding 200^{psi}.
6. If a squeeze pressure develops, hold the pressure for 1 hour before running the drill string

back to bottom.

Notes:

- LCM pills must be thoroughly homogenised and viscous enough before being pumped.
- Pumping high-concentrated LCM pill is subject to use of open-end DP or Mule-Shoe.
- Remove filter screens from the drill pipe (if there is any) and if necessary, remove strainers from the mud pump suction lines.

7.1.2.5 Engineering Comments

Keep watching and control on well to record loss.
All $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.

7.1.2.6 Drilling Issues

There is no drilling objective in AZN-85 workover well; mud should be compatible with reservoir and work over operation.
Solid Control equipment

- Use shale shaker to remove debris and LCM particle in case of use.
- Personal Safety

7.1.2.7 Mud Type

Inhibited CaBr2 brine (Calcium Bromide Brine) will be applied as completion fluid. And simple salt saturated kill mud to kill the well and clean out inside 9 5/8", 7" Liner.

Displacement Design
The combined effect of scrapers being run in the hole, the high flow rate will help to remove/disperse general debris.
At least 150^m of pill/spacer coverage or 3 min contact time at the highest pump rates should be considered.
Saturated $CaCl_2$ brine can be applied as spacer/Hi-Vis pill while displacing.
Pit levels should be monitor throughout the displacement operation to catch and prevent an influx entering the well contributing to an incident.
All $CaBr_2$ brine inside the well must be retrieved by the time of displacing the well by kill mud. In case of need, filtration must be applied to retrieved amount of $CaBr_2$.

No.	Description	Volume [bbl.]	Comment

1	Viscous pill/spacer	50	Clean pill
2	Displace to inhibited CaBr2 Brine at 106pcf (Completion Fluid)		
3	Continue pumping Completion Fluid until cleanliness specification has been met.		

All CaBr2 retrieved from well must be stored in separated tanks. All $CaBr_2$ left on surface, even dead volume of tanks, must be transferred back to drums at end of well. Not even a drop of $CaBr_2$ is not allowed to be wasted or dump in whole operation duration.
It is recommended to use desander tank as active pit of $CaBr_2$ to minimize dead volume and return path.

7.1.2.8 Volume accounting & estimated required material

Work over /Kill Mud Fluid				AZADEGAN #85
Fluid Density (pcf)	106		Fluid System	Water base Work Over Fluid /Salt Saturated
Hole Size (inch)	Completion Phase			
Casing Depth (m)	2903		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	4090
Liner Depth (m)	4090		Liner Lap (m)	2903
Liner OD / ID (inch)	7	6.276	Liner Length (m)	1187

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	350
Cased Hole Volue [Capacity of 3782m 9⅝" Casing + Capacity of 426m 7" Liner]	850
Total New Volume Required (bbl)	**1200**

Estimated Chemical Consumption and Cost based on 106pcf MW							
PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Drill Water	bbl	N/A	0.576 bbl/bbl	700		700	N/A
Lime	25 Kg	N/A	0.3	7	25	32	N/A
Soda Ash	50 Kg	N/A	0.3	4	16	20	N/A
Sodium Chloride	1 MT	N/A	63	37	18	55	N/A

Xanthan-Gum	25 Kg	N/A	0.2	5	20	25	N/A
Calcium Carbonate	1 MT	N/A	325	190	30	220	N/A
Hematite	1.5 MT	N/A	As Needed	As Needed	As Needed	As Needed	N/A

*Above calculation are based on normal hole condition

*Above Calculated is Based on **106pcf**

Completion Fluid			AZADEGAN #85	
Fluid Density (pcf)	106		Fluid System	CaBr2 Brine
Hole Size (inch)	Completion Phase			
Casing Depth (m)	2903		Solids Control Equip Eff.%	
Casing OD/ ID (inch)	9 5/8	8.681	Well TD (m)	4208
Liner Depth (m)	4090		Liner Lap (m)	2903
Liner OD / ID (inch)	7	6.276	Liner Length (m)	1187

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	250
Cased Hole Volue [Capacity of 3782m 9⅝" Casing + Capacity of 426m 7" Liner]	850
Total New Volume Required (bbl)	**1100**

Estimated Chemical Consumption and Cost based on 74-75pcf MW							
PRODUCT	UNIT SIZE	UNIT PRICE (EURO)	CONC'n	Expected Consump. for Mixing	Expected Consump. for Treating	Total Qty Expected To Be Used	TOTAL COST (EURO)
Calcium Bromide (14.2 ppg)	55 Gal	N/A	0.984bbl/bbl	910	62	972	N/A
Corrosion Inhibitor	55 Gal	N/A	1% Vol	9	1	10	N/A
H2S Scavenger Liquid	55 Gal	N/A	0.5% Vol	5	3	8	N/A
Oxygen Scavenger CA	25 Kg	N/A	0.5 ppb	11	6	17	N/A

*Above calculation are based on normal hole condition

*Above Calculated is based on **106^{pcf}**

7.1.2.9 Estimated contingency material

During all the phase keep a minimum safety stock of below items:

Chemical Type	Unit	Qty	Function
$CaCO_3$ M & C	1 MT BB	10 of each	Loss Circulation Material
Pipe-Freeing Agent	55 Gal DM	4	Environment-Friendly Stuck Pipe Agent
Lime	25 Kg Sx	80	CO_2 Contamination
Zinc Carbonate	25 Kg Sx	120	H_2S Scavenger
Sodium Bicarbonate	25 Kg Sx	40	Cement Contamination
Citric Acid	25 Kg Sx	40	Cement Contamination
Defoamer A	55 Gal DM	4	Anti-Foam
Bio-Cide	55 Gal DM	4	Anti-Bacteria
Oyster shell F/M/C	25 Kg Sxs	160each	Loss circulation material
Drilling Detergent	55Gal	4	Bit Balling Reducer
Oxygen Scavenger S	25Kg	30	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	55Gal	4	Polymer Temperature Stabilizer
Corrosion Inhibitor	55Gal	4	Corrosion Inhibitor
Lubricant	55Gal	8	Biodegradable lubricator

7.1.2.10 Material requirements for each well

Item	QTY	Unit	Description	Function
1	972	55 Gal Dr	Calcium Bromide (14.2 ppg)	CaBr2 brine
2	55	1 MT	Salt	Weighting Agent
3	224	1 MT	Calcium Carbonate 75 Mic	Weighting Agent
4	32	25 Kg Sx	Caustic Soda	PH Control
5	20	50 Kg Sx	Soda Ash	Hardness reducer
6	25	25 Kg Sx	Xanthan-Gum	Viscosifier
9	80	25 Kg Sx	Lime	PH Control

10	160	25 Kg Sx	Shell Fish	LCM
11	160	25 KG Sx	Limestone Chips	LCM
12	10	55 Gal Dr	Corrosion Inhibitor	Completion Fluid Corrosion Inhibitor
13	8	55 Gal Dr	H2S SCAV	Completion Fluid H2S SCAV
14	17	25 Kg Sx	OX SCAV	Completion Fluid OX SCAV
15	1	55 Gal Dr	BIO-CIDE	Anti-bacterial
16	12	55 Gal Dr	PH BUFFER	PH Stabilizer

7.2 Cement Program

Not applicable.

8. Perforation Program

Displace 7" Liner volume with $CaBr_2$ perforation fluid, perform 7" Liner perforation, run CCL/GR log to confirm perforated interval. RIH with 5 7/8" Bit to top of perforation while scrap 7" production PKR setting depth, displace well volume with $CaBr_2$ completion fluid. Run 4-1/2" * 3-1/2", 10K Completion string. N/D BOP, N/U XMT. Perform well Pre-Acid clean up, Acid-Wash, Post-Acid clean-up and well performance test operations.

The perforation intervals will be in Fahliyan Formation Fah-2a & Fah-3a with a thickness of 14^m & 4^m TVD respectively. The Contractor must prepare the detail of perforation program under supervision of client.

8.1 Perforation Gun

Wireline conveyed perforation can be the first priority, and TCP (tubing conveyed perforating) to be considered as back-up plan in case of tough conditions in the wellbore.

8.2 Perforation Gun

Perforation gun and charges shall be provided according to Table 9. CCL-GR shall be provided in requested sizes. Guns can be run on wireline or TCP be performed depending on total perforation length.

Gun OD (in)	Shot density (SPF)	Phase angle	API Penetration Depth (in)	Explosive type	Under-balance pressure(psi)
4.5	5	72°	≥40	HMX	NA

Table 9: proposed perforation Gun

9. Completion Program

9.1 Existing Completion string

There is not completion string in the well.

9.2 Proposed Completion string

AZNS-85Completion Sketch

Contractor	MSA	Liner & Tubing Data						Client	
Field	AZADEGAN	Description	7"Liner	7" tie-back 1&2	3 1/2" Tubing	4 1/2" Tubing	Tubing	7" Liner Lap:	3829
Well Number:	AZNS-85	OD[in]	7.00	7/00	3.50	4.50		7" Liner Shoe:	4090
Rig :	OEOC 201	Nominal weight [lb/ft]	35.00	29.00	10.20	18.90		7" Tie-Back L.Lap #1:	3012
RTE-GLE	3.95	Grade	VM110HC-SS	L-80	L-80	L-80		7" Tie-Back L.Lap #2:	2903
RTE-THF	13.45	Thread	N.VAM	HSM1/VAM	HSM3	HSM3			
Rig Spudded Date :		Inside diameter [in]	6.004	6.063	2.92	3.640		Total Depth:	4090
End of Operation		API Drift Diameter	5.879	6.059	2.80	3.515		X-mass Tree:	-
								Size & Rating:	

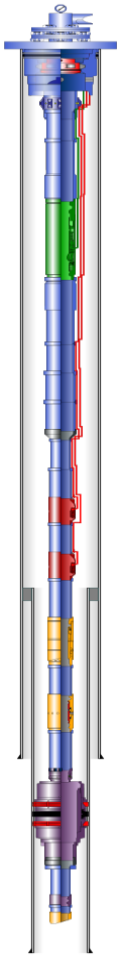
Schematic		Description	Max OD	Min ID	Length(m)	BTM(m)	TOP(m)
	1	Elevation (floor - TBG Head Flange)			9.5		
	2	TBG HANGER 11" * 4 1/2" ACME Box up * 4 1/2" TPCQ-EX Box Down , M/C: FF-0.5 T/C: U PSL: 3G PR: 2, 4" Type "H" Two Way	11	3.68	0.65	0.650	0.000
	3	4-1/2" FLUTED SWAGE ADAPTER D/P, 18.9# TPCQ-EX PIN ON TOP X HSM3 PIN ON DOWN* SUPER 13CR-110KSI	5.5	3.64	1.04	1.690	0.650
	4	5 JOINTs TUBING 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	47.5	49.190	1.690
	5	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	50.810	49.190
	6	4-1/2" FLOW COUPLING,18.9#,SUPER 13CR-110KSI,HSM3 , 6 FT	5.5	3.64	1.72	52.530	50.810
	7	4-1/2" X 3.437" TRSSV-10 TUBING RETRIEVABLE SURFACE CONTROLLED SAFETY VALVE WITH SELF-EQUALIZING FEATURE, WITH AF PROFILE	7.5	3.437	1.78	54.310	52.530
	8	4-1/2" FLOW COUPLING, 18.9#,SUPER 13CR-110KSI,HSM3 , 4' FT	5.5	3.64	1.04	55.350	54.310
	9	PUP JOINT 4 1/2",HSM3 ,18.9#, L-80, Type-1	5.118	3.64	1.62	56.970	55.350
	10	289 JOINTs TUBING 4-1/2"HSM3 18.9#, L-80	5.118	3.64	2750.702	2807.672	56.970
	11	X-over 4-1/2"HSM3, 18.9# BOX * 3-1/2"HSM3 , 10.2# PIN,	5.118	2.922	1	2808.672	2807.672
	12	9 JOINTs TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	85.5	2894.172	2808.672
	13	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2895.992	2894.172
	14	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	2896.722	2895.992
	15	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2898.542	2896.722
	16	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	2908.042	2898.542
	17	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2909.862	2908.042
	18	3 1/2" CIV TUBING RETRIEVABLE CHEMICAL INJECTION MANDRELHSM3 , 10.2#,SUPER 13CR-110KSI W/CHECK VALVE, DUAL TYPE & RUPTURE DISC	5	2.922	0.73	2910.592	2909.862
	19	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.8	2912.392	2910.592
	20	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	2921.892	2912.392
	21	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2923.712	2921.892
	22	3-1/2" EXP. JOINT,HSM3 , SUPER 13CR-110KSI, 10'STK (Half Closed) Splind Rotational Lock System Any Position, Shear Pin Set 20,000 LBS,	5.315	2.922	5.755	2929.467	2923.712
	23	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2931.287	2929.467
	24	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	2940.787	2931.287
	25	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.62	2942.407	2940.787
	26	3-1/2" NON-ELASTOMERIC SLIDING SIDE DOOR, SUPER 13CR-110KSI, HSM3 With Type X Selective Profile (PROFILE: 2.750" X, DOWNWARD OPEN)	4.551	2.75	1.19	2943.597	2942.407
	27	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.17	2944.767	2943.597
	28	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	2954.267	2944.767
	29	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.72	2955.987	2954.267
	30	3-1/2" ANCHOR SEAL ASSEMBLY,HSM3, 10.2#,SUPER 13CR-110KSI, 4" PACKING OD	4.00	2.922	0.23	2956.217	2955.987
	31	7" HYDRAULIC SET PERMANENT PRODUCTION PACKER, MODEL WSDP TO BE RUN IN 7.00" 29-32 PPF CASING, SUPER 13CR-110KSI, UPPER SEAL BORE: 4"	5.91	2.922	4.4	2960.617	2956.217
	33	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2962.437	2960.617
	34	1 JOINT TUBING 3-1/2"HSM3 , 10.2#, L-80.Type-1	4.25	2.922	9.5	2971.937	2962.437
	35	3-1/2" PUP JOINT,HSM3 , 10.2#, L-80, Type-1	4.25	2.922	1.82	2973.757	2971.937
	36	3-1/2" BOTTOM NO-GO LANDING,HSM3 , 10.2#, SUPER 13CR-110KSI W/ Seal Bore 2.75" TYPE XN	4.5	2.635	0.37	2974.127	2973.757
	37	3-1/2" WIRLINE RE-ENTRY GUIDE, 10.2 #,HSM3 Con,SUPER 13CR-110KSI	4.25	2.922	0.24	2974.367	2974.127
		END OF COMPLETION STRING				2974.607	2974.367
							2974.607
Tubing Retrievable Saftey Valve						54.310	52.530
Traveling Joint						2929.467	2923.712
CIV 1						2896.722	2895.992
CIV 2						2910.592	2909.862
SSD						2943.597	2942.407
7" HYDRAULIC SET PERMANENT PRODUCTION PACKER						2960.617	2956.217
NO-GO Landing Nipple						2974.127	2973.757
Entry Guide						2974.367	2974.127

Figure 7: Proposed Completion Schematic

10. BOP, Wellhead & X-Mass Tree

10.1 Existing Wellhead & X-Mass Tree



Figure 8: Existing Wellhead

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Savark	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K

Item	Specification
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Savark & water)
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Savark & Water disposal wells)
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)

Table 10: AZNS Wellhead Information

- **Wellhead Specification**

NOTE: Wellhead & X-Mass Tree already installed by client.

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports & seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange & 2" bull plug threaded for ½" NPT & fitted with 0-3,000 psi gauge & needle valve &, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange & 2" bull plug. Casing slip & seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** & bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space & casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange & 2" bull plug on one side &, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT & fitted with 0-5,000psi gauge & needle valve. Casing slip & seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130
Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** & bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space & casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT & fitted with 0-10,000psi gauge & needle valve on one side &, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT & fitted with 0-10,000psi gauge & needle valve, & the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange & 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck & two S-SEAL to pack off the body of hanger, & tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm**

	Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) & with test ports to test annular space & seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG & two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge & needle valve installed & one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch & with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve & one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top, including 1/2"NPT port for needle valve & pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor & gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

Table 10: AZNs 11 Wellhead Specification

10.2 BOP

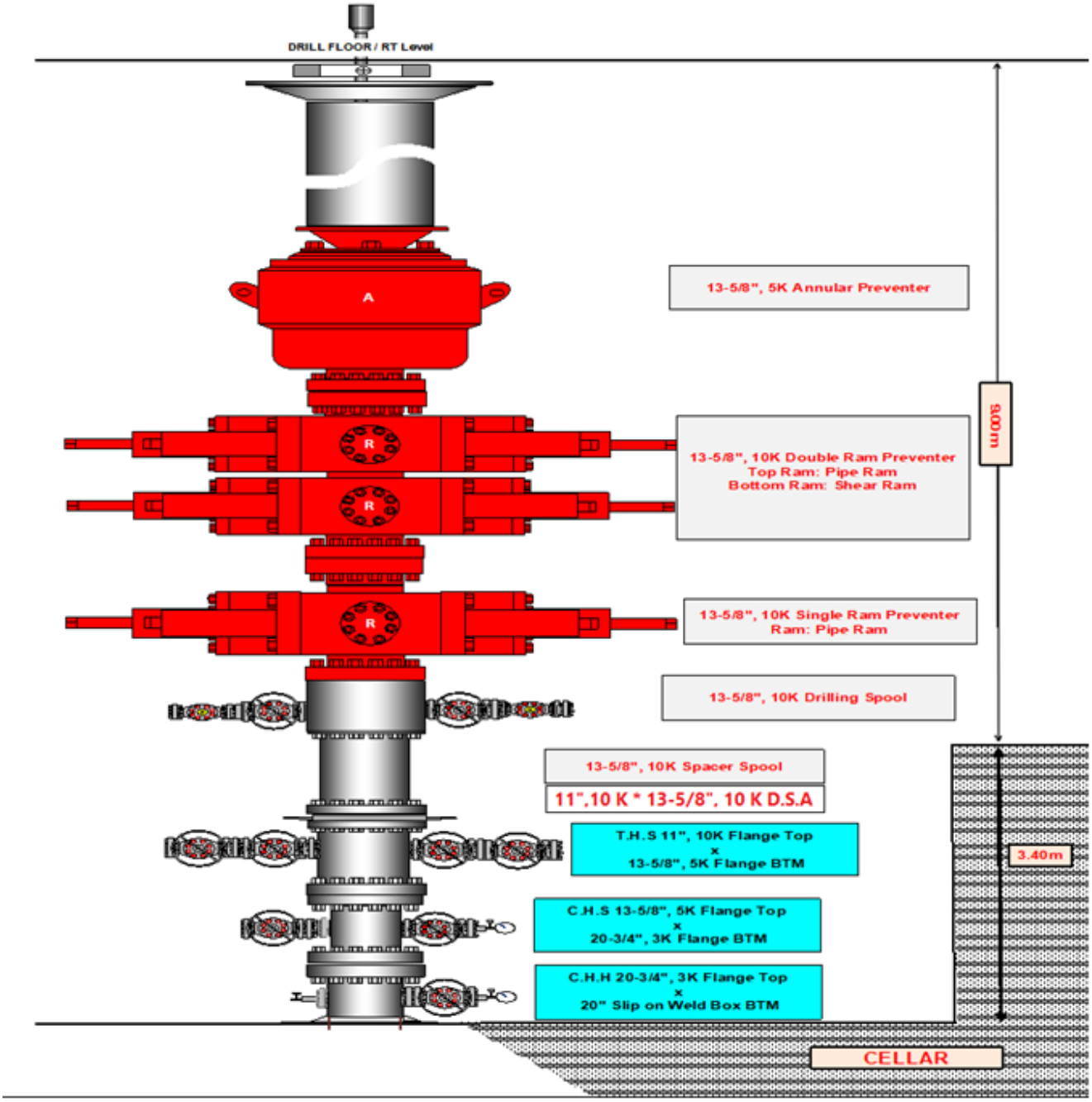


Figure 9: BOP Sketch

11.

11. Waste Management

There is no drilling objective & drilling cutting in AZNS-85 workover well; therefore waste management program is not applicable.

In case of requirement to dispose wastewater in workover well, waste management program will be issued at due time as PEDEC policy.

12. Health, Safety & Environments

12.1 WELL CONTROL

All the drilling operation shall obey the requirement of Well Control Manual of MSA & contractor's well control manual; under no circumstances should the deviation from well control policy be permitted without prior approval through MSA. If the contractor's well control manual or anywhere has conflict with Well Control Manual of MSA, take Well Control Manual of MSA as the standard.

12.1.1 WORK Requirement of Well Control Equipment

- With shut-in casing pressure lower than 1,000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, & the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2,000psi, drill tools are allowed to pull up & down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone & thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.
- All gate valves of well control manifolds shall be tagged with serial numbers, & be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole & close pipe rams on it to prevent the possibility of shear ram's failure.

12.1.2 Management of Well Control Equipment

- Well control equipment used by the drilling crew shall be managed & operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom & rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components & rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals & ram rubber, every type gasket & bolt etc. should be enough for backup.

12.2 Recording Reference

All BOP tests, drills, function tests, any malfunctions, repair or maintenance to the mud System & well control equipment shall be recorded in the IADC daily reports & shall be signed by both the Drilling Contractor's Tool Pusher & Drilling Supervisor on the well site. They shall also be recorded in the Daily Drilling Report.

12.3 Flow Check Procedure

To avoid a kick situation, it is recommended that a flow check be conducted at specific times during the drilling operation.

While drilling (any warning sign).
Prior to tripping the drill string.
While tripping (any warning sign).
While out of hole (continuously).
Trip out:
After pulling 5% of the drill pipe
At the midpoint depth of the wellbore
Prior to pulling the first stand of drill collars
After all the drill string is out of hole.
Trip in:
Upon reaching surface casing shoe
At midpoint of the wellbore

12.4 Primary Control Reference

It is the responsibility of the Drilling Contractor to keep the hole full at all times by using all the available equipment. The Drilling Supervisor shall monitor that the correct preventive practices are being implemented and/or conducted at all times.

Extreme care shall be taken to monitor mud volume, drilling breaks & gas cut mud.

The MACP (Maximum allowable casing pressure) shall be confirmed by the Drilling Supervisor, either after setting each new casing string or, whenever the density of the drilling mud changes.

The MACP shall be clearly written on a kick control sheet that will be posted near the choke control panel.

The predetermined distance at which the Tool Joint is to be pulled above the rotary should be posted on rig floor near the BOP control panel.

The Driller is responsible for carefully measuring & recording the slow pump pressure.
Awareness of these values is an important element in killing operations, in order to avoid formation breakdown.

Take a measure & record the slow pump pressure at three different flow rates with all pumps at the following times:

Once per tour.

Whenever changes occur in the mud density/rheology.

Whenever changes occur in the dimension & characteristics of the string, BHA, jet size, jet plugged or jet lost, etc.

12.5 Secondary Control Reference

Any time a drilling break is noticed, drilling shall stop & a static monitoring of the well shall be carried out. Space-out for all Pipe Rams & Annular Preventer must be known to the Driller at all times to keep tool joints clear of annular preventer & rams.

12.6 Well control during drilling & completion operation

The drilling design shall be strictly implemented by the drilling contractor & when it is discovered there is incompliance of downhole situation with design & the design will be amended, the procedure for amendment of design shall be followed. If emergency is encountered, & safe operation can be affected, after the discussion between drilling rig manager & the drilling supervisor, practical technical measures can be taken first & then report to the top management. The Company shall approve & notice in time as per the drilling program amendment procedure.

The curve of well equivalent mud weight as per predicted formation pressure, curve of mud weight in drilling program & curve of actual mud weight must be drawn during drilling operation. For exploratory well & key well, the curve of dc index while drilling formation pressure equivalent mud weight for sandstone, mudstone & shale shall be drawn by the comprehensive mud logging crew.

When entering oil & gas formation or drilling master reservoir, high diameter nozzle shall be selected for the drilling bit to treat well killing & mud loss.

SCR (Slow Circulation Rate) shall be conducted at the start of drilling activity, pump strokes, pumping rate & circulating pressure should be recorded. While drilling in pay zone, SCR is required to be conducted with all pumps separately as follows, pump strokes, pumping rate & circulating pressure shall be recorded in Daily Drilling Report & Kill Sheet.

Once for every 12 hours in general;

After bit change or BHA change; 3) Mud weight change exceeds 3.75pcf.

Oil & gas invasion & overflow shall be checked with wiper trip & shall be conducted under the following situation:

Each time before POOH after drilling oil & gas formations.

Severe invasion of oil & gas occurs during drilling, but overflow does not occur before POOH.

POOH after killing well.

POOH after lowering mud weight in wellbore.

POOH after curing mud loss during drilling oil & gas formation.

POOH after drilling oil & gas formation & a prolonged period of time is needed for other operations (E logging, running casing, running tubing & DST etc.).

Regarding the shut-in methods selection

Soft shut-in method is recommended for gas well;

Hard shut-in method is recommended for oil well;

Hard shut-in method is recommended for the well of which the well has strong formation strength in EMW as per FIT test;

Soft shut-in method is recommended for the well of which the well has weak formation strength in EMW as per FIT test.

Main technical measures for prevention of kicking and blowout during tripping.

Before POOH, circulation duration of mud in wellbore shall not be less than one and a half cycles. The observation time after wiper trip shall be no less than one and a half cycles. The density differential of inlet and outlet shall not exceed 1.248 pcf.

Flow check shall be conducted regularly via trip tank as follows

Prior to POOH (keep string rotation at 10 rpm for preventing stuck pipe)

POOH to last casing shoe

Before POOH BHA through BOP stack

Note: Tripping should be shut-off immediately and flow check shall be performed later on when any suspected well kick is observed.

During POOH, fill up hole every 3-5 stands of DP and every stand of DC. If the nozzle is plugged, during POOH, mud shall be filled full for each joint or continuous filling shall be conducted and trip sheet shall be recorded carefully.

When the bit is in the oil and gas formation and 300m above top of oil and gas formation, POOH speed shall be controlled to maintain the excellent well building property and rheological property of mud to avoid severe swabbing in the wellbore.

During RIH, the running speed shall be controlled to avoid the loss of circulation due to down hole pressure surge.

During tripping, check and repair of the equipment shall be arranged at the time when running to the casing shoe. If equipment have to be checked and repaired during POOH, due blowout prevention measures shall be taken during and after the check and repair, drill string shall be run to bottom and circulate for one and a half cycles. After everything is normal, drilling shall be resumed. Equipment check and repair shall not be conducted for empty well condition.

Treatment measures for invasion of gas during normal drilling:

When invasion of gas is discovered, it shall be discharged with circulation in time. The gas invaded mud shall not be pumped in the hole before eliminating gas.

Increasing mud weight shall be conducted when stop drilling and after invaded gas eliminated from mud. Mud weight incremental of each circulation cycle shall be controlled within 3.121 pcf

"In-post" policy

"In-post" will be initiated after the surface casing is cemented.

The "in-post" personnel must be trained and verified qualified by the technical staff of drilling rig.

During drilling, specific personnel shall be arranged to be "in-post", the mud logging personnel shall monitor the level of the circulating tank etc. as per the requirements. During tripping, other auxiliary operation or drilling suspension, specific personnel for "in-post" shall be arranged by the drilling and mud logging personnel in the meantime.

"In-post" requirements:

For drilling operation, variation of the outlet flow, drilling time, lithology, bubble, smell, showings, pit volume, density and mud viscosity, gas measured values and content of chloride etc. shall be carefully monitored. The level of the circulating tank shall be observed and recorded each 15 minutes and if special situation occurs, the measurement and record shall be intensified. If anomaly occurs, it shall be reported to the driller in time.

For tripping operation, trip sheet shall be filled accordingly. If anomaly occurs, it shall be reported to the driller.

For other operations, it shall be observed whether there is overflow of mud at the wellhead or mud level from wellhead. If anomaly occurs, it shall be reported to the driller.

Treatment of loss of circulation

Before cementing non-target zone or pay zone, if loss of circulation occurs during drilling, POOH shall be carried to run open ended drill pipe for cement plug job if condition allows. During RIH, mud shall be filled from annulus to balance formation pressure to avoid blowout.

Treatment of loss of circulation during drilling after production casing cemented

Mud level not at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and observe with moving the drill string and prepare well control related work. Each 10 minutes, mud with density no lower than the drilling weight shall be reversely filled up from the annulus. Each time, the filling up shall be 1~2m3, or continuous reserve injecting mud with small speed to maintain that the column pressure of mud in wellbore is higher than the formation pressure. POOH to casing shoe or

safety interval for observation. Additional treatment shall be made as per the situation of loss of circulation.

Mud level at wellhead after loss of circulation. Stop drilling, pick up TDS or Kelly, stop pumps and move the drill string to observe with well control well prepared. When the well is static, pull the bit upward to the casing shoe or safe interval for observation. If no anomaly is observed, circulate with slow pump rate. The measures such as plugging back job, POOH or condition mud etc. shall be carried out as per the situation of loss of circulation. If loss of circulation occurs to pay zone, mud weight shall be acquired by static pressure. After pulling out to casing shoe and observing for one tripping cycle, when no more loss of circulation occurs after running to bottom, drilling operation can be resumed.

Well control for E logging, cementing and DST

Before E logging, down hole condition shall be stable and static. If it takes long time for electrical logging, one more condition trip shall be conducted between logging. Shooting nipple is required for logging. Logging service shall be furnished with tools for cutting the cables.

Continuously fill the well during performing wireline logging, and keep closely monitoring for the well via trip tank.

Before running casing, ensure the well is in static condition without any flowing. Pipe ram shall be changed to casing ram with proper size. Casing circulating head or crossover sub between casing and in used DP shall be kept at rig floor at all times during casing running.

The hydrostatic pressure shall be kept balance with formation pressure during whole cementing job to avoid blowout due to loss of circulation, loss of balance for pressure and cement weight loss during WOC.

12.7 HEALTH

This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.

A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.

An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centres manned by qualified doctor 24-hours per day in Ahwaz city.

Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.

12.8 SAFETY

Due to high concentration of H2S in Sarvak, Kazhdumi and Gadvan reservoirs, Make detail H2S operation procedures, in particular, and carry it out strictly.

Lifesaving equipment like SCBA, Escape Mask, H2S detectors (Multi portable and personal detector) and implement the H2S Services system during drilling activity should be provided.

All lifting equipment and cranes must be certified by approved 3rd party (approved by Institute of Standards & Industrial Research of Iran)

All electrical equipment should be designed for such locations, and listed by nationally recognized testing laboratory. All wiring components and electrical equipment should be maintained in accordance with the manufacturers' recommendation.

PSSR (Pre-Startup Safety Review) before spud-in and penetrating into oil and gas formations are mandatory by contractor under the supervision of Company and approved by Company.

12.9 ENVIRONMENT

Requirement on location

The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.

Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.

Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.

Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.

Contractor shall perform on-site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation Contractor in accordance with the following waste hazard classification.

Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)

Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass.

Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.

The Contractor/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.

It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to.

Local community are recommended.

Hazardous Wastes
Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site
Mud purifying circulation system.

Cooling water spraying and circulating system for drilling pump.

Special recovery tank for deposed oil.

Reserve house for valuable mud chemicals.

Automatic mud weighting supply system.

BOP stack is equipped according to drilling design.

Define a separation area for gathering all waste according to waste management procedure

Requirement of environment protection on location
Sewage pit Area of the sewage pit: 1500 m2.
Prior to spud, mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.

Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.

Install oil supply lines and check whether there is any leakage.

Management of environment protection on location

In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.

Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.

Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.

Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.

After the well is completed, reinforce the sewage pit solidify the disposed mud and have environment procedure fulfilled.

Install fuel supply lines and keep them in good condition.

All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.

The environmental protection personnel from contractor and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.

Transition Area for Hazardous Waste shall also be provided by each Contractor in his allocated area, Company shall approve each area before use as well. The transition area of hazardous waste shall be:

Paved, fenced, marked and illuminated

Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g. Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).

In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).

Configured to accommodate a safe physical separation of incompatible waste materials (e.g. acids, and alkalis, organic materials and strong oxidizing agents).

Capable enough to provide storage for the entire range of hazardous wastes generated during the work.

The area shall be provided with Spill Containment and means of preventing run-off to and from the area.

All waste containers to be stored shall be appropriately labelled.

Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from Company and/or PEDEC.

13. H_2S Monitoring & Prevention Measure

13.1 Monitoring H2S

Somewhere on the well site is apt to accumulate H2S, such as mud pit, shale shaker, and drilling platform, should be installed one set of. Make sure H2S monitor and alarm system can be used. Alarm threshold setting for H2S warning indicator: the first warning level is 10mg/m3, unable warning sound, just show present value of H2S content to operators. The second warning level is 20mg/m3, when the H2S content in air exceed 20mg/m3, H2S warning indicator can give an alarm automatically, and warning sound should be heard by everyone work in well site. The audible alarm should be installed on Kelly board.

Equip enough gas masks on drilling platform, shaker screen (total amount should large than 24), and equip 12 sets of 30min positive pressure Air Aspirators 2 sets of portable H2S monitor and appropriative H2S warning indicator. Arrange a person specially assigned for taking charge all the protective apparatus. This man must do periodical check to keep these equipment in a good condition.

H2S monitor should be done weekly check and compulsory examination.

After drilling into the pay zone, strengthen the detection of H2S content in mud.

13.2 Measures for H2S prevention

Windsocks shall be set on the drill floor and in the monsoon direction or/and prevailing wind direction of Azadegan oil field, off well site. In emergency, employees must follow the evacuation procedure.

Explosion-proof blowers shall be installed around the rig, shale shaker, and drilling fluid tanks to blow away toxic and flammable gas at the well site.

The rig crew in the H2S area shall be furnished with H2S monitors and adequate protective apparatus. Everybody shall be able to use, maintain, and check the apparatus.

The technicians of rig crew are responsible for H2S proof training and the tool pusher is in charge of the supervision and examination.

The pH value of drilling fluid in the H2S area shall be kept over 9.5. Pay attention to measurement of H2S concentration in drilling fluid. The H2S concentration shall be kept at less than 50mg/cm3 by using sulphur removal agents and degassers.

Personnel safety and protection shall be carried out according to SY/T5087 during operations in polluted areas where H2S content exceeds the critical concentration.

When the rig crew makes decision to bleed and fire because well control cannot be performed due to well site conditions, operations shall be performed according to procedures in SY/T5087.

Once blowout occurs, the incident shall be reported to the higher level department as soon as possible and firefighting trucks, ambulances, medical care personnel and technicians shall be on duty at the well site.

After the blowout is controlled, H2S concentration should be measured in the places where H2S is possible to gather.

The personnel are not allowed to access until the H2S concentration is below the critical concentration.

14. Handover Data

The handover document should contain the following well construction information:

- Pre-spud Inspection Report
- Daily Drilling Report
- Daily Mud Report
- Daily Mud Logging Report
- Perforation Data
- Completion Tally
- Completion running Report
- TMS data (Completion)
- Dry Test report
- Test Charts, X-Mass tree Installation & test Report.

15. Requirements

Item	Description	QTY.	Unit	Supplier
1	8-1/2" Milled Tooth Bit	1	ea.	MSA
2	5-7/8" Milled Tooth Bit	1	ea.	MSA
3	5-7/8" Mill	1	Ea.	MSA
4	4 1/2" * 3 1/2" Completion String 10K	1	set	MSA
5	9-5/8" & 7" Scraper	1+1	ea.	OEOC
6	5" DP,HWDP,8-1/2" DC,	as required		OEOC
7	Well Head Services	as required		Service Co.
8	Surface Well Test Services	as required		Service Co.
9	Completion Services	as required		Service Co.
10	Slick line Services	as required		Service Co.
11	Casing Running Services	as required		Service Co.

12	Wireline Logging	as required		Service Co.
13	Dry Test, DST, RTTS	as required		Service Co.
14	Cementing Services	as required		Service Co.
15	Stimulation (Coiled Tubing, Acidizing and N2) Services	as required		Service Co.
16	H2S Services	as required		Service Co.
17	Mud Logging Services	as required		Service Co.
18	Rig Services	as required		Service Co.
19	Drilling & Completion Fluid Services	as required		Service Co.

Note: Quantity & Spec to be conformed with contract's SOW.
Note: In case of any shortage or discrepancy, the contract's SOW will be reference.

Table 12: List of Requirements

16. Revision Corrections

Drilling Engineer							
1	**برنامه ارائه شده مطابق با شرح کار تایید شده کارفرما نمی باشد.** مطابق با شرح کار تایید شده، می بایست یکپارچگی آستری ۷ اینچ محرز گردد. **Well information:** • مدت زمان پیش بینی شده مطابق با SOW تایید شده نمی باشد.	8	CM	• مطابق مکاتبات فی مابین در انتظار اخذ تاییدیه کارفرما می باشد. • با توجه به تاریخچه چاه تنها امکان نشتی سوزنی از آستری پس آویز اول در عمق ۳۰۱۲ متری وجود دارد لذا پیرو نامه شماره ۱۴۶۷۹۱ مورخ ۱۴۰۲/۱۰/۱۹ و سناریو های مطرح شده راندن آستری ۵ اینچ با ریسک بالا همراه است و ممکن است منجر به از دست دادن چاه گردد. Contingency با توجه به اجرایی نبودن تغییرات نسبت به شرح کار Plan #1-2 ایجاد شده است. • نام جدول تصحیح گردید.	• طبق قرار داد پیمانکار موظف به انجام شرح تعهدات خود می باشد لذا اجرا نکردن عملیات مذکور منوط به تایید کارفرما می باشد. • برنامه ارائه شده توسط پیمانکار یکپارچگی آستری ۷ اینچ را **ارزیابی نمی نماید** و همچنین راهکار ارائه شده از سوی پیمانکار مبنی بر تغییر نصب عمق پکر (حدود ۱۰۰۰ متر بالاتر) بدون احراز یکپارچگی آستری ۷ اینچ مورد **تایید نیست.** • برنامه ارائه شده از سوی پیمانکار می بایست عملیات را مطابق با شرح کار تایید شده، دنبال نماید. • نام جدول تصحیح گردد. صفحه ۸		CM
2	**Time Breakdown:** • با توجه به SOW مدت زمان پیش بینی شده می بایست	23-24	CM	• مطابق مکاتبات فی مابین در انتظار	• .مطابق آیتم اول		CM

	اصلاح شود. • **با توجه به مغایرت contingency plan با SOW می بایست Contingency plan در برنامه به صورت کامل قید گردد.**			اخذ تاییدیه کارفرما می باشد.			
3	**Sequence of Operation:** • **با توجه به مغایرت contingency plan با SOW می بایست Contingency plan در برنامه به صورت کامل قید گردد.**	25-26	CM	• مطابق مکاتبات فی مابین در انتظار اخذ تاییدیه کارفرما می باشد.	• مطابق آیتم اول		
4	گل مورد استفاده جهت حفر پلاگ سیمانی می بایست فشار هیدرواستاتیک بیشتر از فشار سناریوی ورود اینفلاکس گازی زیر پلاگ سیمانی را داشته باشد و همچنین تمهیدات لازم جهت رخداد شرایط مذکور مد نظر پیمانکار قرار گیرد.		CM				**AP**
Fluid							
5	• به دلیل احتمال جریان چاه و آلودگی نفتی، شرکت عملیات اکتشاف باید تمهیدات لازم جهت تمیز سازی محوطه و جلوگیری از نشت مواد آلوده نفتی به محیط اطراف، را در نظر بگیرد (با استقرار وسایل نقلیه حمل سیال و امکان بلوک چینی).		CM				
Completion & Production							
6	• Packer depth مربوط به Completion بعد از برسی Corrosion Log نهایی می گردد.		CM				**AP**

7	• با توجه به بالاتر آمدن رشته تکمیلی (حدود ۱۰۰۰ متر) وتغییرقطر مسیر جریان در بازه مذکور، تغییرات فشارسرچاه و میزان نرخ تولید مربوط به آن می بایست بررسی گردد.		CM	• بر اساس اطلاعات دریافتی از کارفرمای محترم و شبیه سازی انجام شده بالاتر آمدن عمق پکر تغییر محسوسی در شرایط نرخ تولید ایجاد نمی کند.	به پیوست ارائه گردیده است.		CM
8	• با توجه به اینکه در صورت نصب پکر در عمق پیشنهادی جدید (تقریبا هزارمتر بالاترازعمق قبلی)، انتهای رشته تکمیلی و در نتیجه عمق شیر تزریق افزایه های شیمیایی(CIV) نیز تغییر خواهد کرد و تقریباً هزار متر بالاتر خواهد رفت.در صورتی که امکان ایجاد آسفالتین در این بازه۱۰۰۰متری وجود داشته باشد دچارمشکل خواهد شد لذا عمق شیر تزریق در محدوده ای باشد که افت فشار در حدی است که رسوب آسفالتین تشکیل نشده است؛ - - بنابراین فشار مربوط به نقطه شروع رسوب آسفالتین را باید با بازه فشار در محدوده ۱۰۰۰ متری که لوله حذف می شود مقایسه کرد. **موارد فوق در خصوص تولید می بایست بررسی و مد نظر قرار گیرد.**			• بر اساس اطلاعات دریافتی از کارفرمای محترم (فشار تشکیل آسفالتین 5500 psi) عمق نصب شیر های تزریق افزایه های شیمیایی بر اساس عمق پکر (۲۹۵۶ متری) مناسب می باشد.	به پیوست ارائه گردیده است.		CM
Reservoir							

CM	37	• جدول 7.1.2.8 کپی گردیده AZNS122اطلاعات از چاه بنابراین اطلاعات بازبینی گردد. و نام چاه به ۰۸۵ تغییر کند.	• اصلاح گردید.		36 AND 37	7.1.2.8جدول کپی گردیده بنابراین AZNS122اطلاعات از چاه اطلاعات بازبینی گردد. و نام چاه به ۰۸۵ تغییر کند.	9
Geology							
CM		• انجام نشده است	• با توجه به اینکه جدول شماره ۴ شامل اطلاعات مخزنی، قبلا مطابق با Comment sheet-00 اصلاح گردیده بود. جدول مورد نظر به صورت مجزا با عنوان آیتم ۱-۶ در صفحه ۱۶ گنجانده شد.	CM	14	مقادیر ذکر شده با شرح Reservoir Data صفحه ۱۴ جدول ۴ کار متفاوت است پیشنهاد می شود همان جدول شرح کار اورده شود	10
CM		• سرسازند فهلیان با شرح کار ختلاف دارد	• اصلاح گردید.	CM	18	• صفحه ۱۸ شکل ۴ عمق تاپ پلاگ سیمانی ۲ ۲۷۶۳ متری ذکر شده در صورتیکه عمق صحیح ۲۷۶۸ متری می باشد. • صفحه ۱۸ شکل ۴ عمق سر سازندها با شرح کار مغایر است.	11
CM		• مطابق آیتم اول	• مطابق مکاتبات فی مابین در انتظار اخذ تاییدیه کارفرما می باشد.	CM		، راندن لاینر ۵ Contingency Planدر شرح کار تایید شده اینچ پیش بینی شده است. در صورتیکه در برنامه چاه ۸۵ اشاره ای به آن نشده است. **در صورت راندن لاینر ۵ اینچ برنامه مشبک کاری طبق جدول زیر خواهد بود که در برنامه ارسالی گنجانیده شود.**	12

Table 7: proposed perforation Gun for 5" liner

Gun OD (in)	Shot density (SPF)	Phase angle	API Penetration Depth (in)	Explosive type	Under-balance pressure(psi)
3 3/8	6	60°	≥30	HMX	NA